

MEDLEY: Medication Embeddings for Longitudinal Phenotyping

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Motivation

Medication use evolves in response to changing patient state

- Accurate phenotyping of longitudinal medication trajectories is critical for risk stratification and targeted interventions

EHR medication data present major modeling challenges

- Irregular, patient-specific administration timing
- Repeated dose adjustments and drug switching
- Multiple routes and formulations with non-uniform units
- Inconsistent generic and brand naming

Traditional feature engineering is insufficient

- Binary indicators ignore duration and intensity
- Mean doses obscure drug escalation and switching patterns

How can we flexibly represent medication trajectories from tabular EHRs while preserving their temporal dynamics?

Background

Transformers leverage self-attention to capture long-range dependencies in medication sequences

- Self-attention adaptively weighs relationships between events without requiring fixed time intervals
- Several transformer-based large language models (LLMs) have been pretrained on biomedical literature and clinical notes [1, 2]
- Encoder LLMs produce high-dimensional latent representations known as embeddings; decoders generate new text

Methods

1) Template construction

- Convert tabular EHR medication data into sequential text representations
- List/Text templates: Manual transformation of medication records [3]
- LLM Template: Decoder-only LLM (Llama-3.3-70B-Instruct) generates concise summaries of List template

2) Embedding generation

- Encode medication sequences using long-context encoder-only LLMs: Clinical-Longformer (CL; clinically pretrained), GatorTron-Base-2k (clinically pretrained), and Longformer (LF; general-domain comparison)
- Apply PCA for dimensionality reduction prior to clustering

3) Unsupervised phenotype discovery

- Cluster embeddings using K-means++, Gaussian mixture models (GMM), HDBSCAN, and agglomerative hierarchical clustering (AHC)
- Optimal number of clusters selected using elbow methods
- 50 repetitions for non-deterministic algorithms
- Internal validation metrics: Silhouette score (SS $\uparrow$), Calinski–Harabasz index (CHI $\uparrow$), Davies–Bouldin index (DBI $\downarrow$)

Application: Postoperative delirium in MIMIC-IV

Postoperative delirium (POD)

- Common postoperative complication in older adults characterized by inattention and fluctuating mental status
- POD management involves multiple sedative and antipsychotic medications, often with dynamic dose adjustments and drug switching

Study cohort

- 5,821 ICU surgical patients with POD from the MIMIC-IV database [4]

MEDLEY Framework

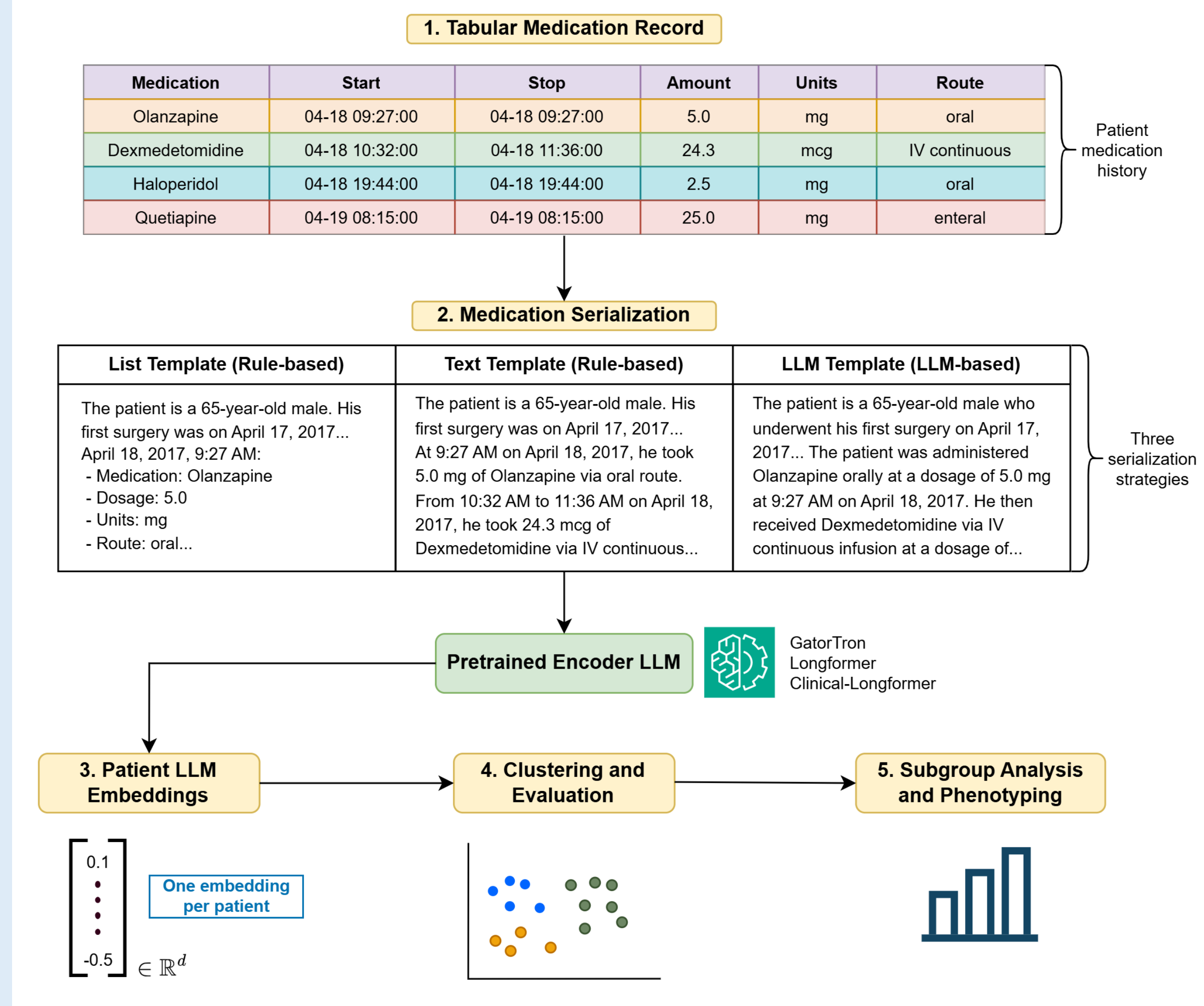


Figure 1. Overview of the MEDLEY framework

Embedding and Clustering Performance

Table 1. Internal validation metrics across embedding templates and clustering algorithms

Algorithm	Template	SS ($\uparrow$)	CHI ($\uparrow$)	DBI ($\downarrow$)	Number of Clusters
K-means++	CL List	0.785 $\pm$ 0.068	1.43e5 $\pm$ 2.23e4	0.457 $\pm$ 0.065	5
	CL Text	0.776 $\pm$ 0.052	8.85e4 $\pm$ 6.49e3	0.474 $\pm$ 0.048	5
	CL LLM	0.665 $\pm$ 0.040	3.61e4 $\pm$ 2.63e3	0.505 $\pm$ 0.007	5
	GT-base-2k List	0.741 $\pm$ 0.050	7.17e4 $\pm$ 6.32e3	0.503 $\pm$ 0.032	5
	GT-base-2k Text	0.730 $\pm$ 0.000	5.99e4 $\pm$ 1.38e-2	0.504 $\pm$ 0.000	5
	GT-base-2k LLM	0.608 $\pm$ 0.055	2.85e4 $\pm$ 2.02e3	0.584 $\pm$ 0.026	5
	LF List	0.761 $\pm$ 0.048	1.03e5 $\pm$ 2.40e4	0.460 $\pm$ 0.045	5
	LF Text	0.756 $\pm$ 0.020	7.07e4 $\pm$ 5e3	0.481 $\pm$ 0.043	5
GMM	LF LLM	0.691 $\pm$ 0.046	4.29e4 $\pm$ 4.41e3	0.485 $\pm$ 0.020	5
	CL List	0.720 $\pm$ 0.035	1.28e5 $\pm$ 1.83e4	0.519 $\pm$ 0.024	6
	CL Text	0.681 $\pm$ 0.022	7.04e4 $\pm$ 1.05e4	0.647 $\pm$ 0.055	6
	CL LLM	0.548 $\pm$ 0.008	2.60e4 $\pm$ 3.53e3	0.690 $\pm$ 0.199	6
	GT-base-2k List	0.568 $\pm$ 0.159	3.59e4 $\pm$ 5.18e3	0.771 $\pm$ 0.160	6
	GT-base-2k Text	0.683 $\pm$ 0.001	4.98e4 $\pm$ 3.14e2	0.512 $\pm$ 0.002	6
	GT-base-2k LLM	0.453 $\pm$ 0.038	2.01e4 $\pm$ 4.83e2	0.784 $\pm$ 0.083	6
	LF List	0.700 $\pm$ 0.021	1.19e5 $\pm$ 1.30e4	0.581 $\pm$ 0.134	6
HDBSCAN	LF Text	0.594 $\pm$ 0.064	4.07e4 $\pm$ 1.71e4	0.854 $\pm$ 0.343	6
	LF LLM	0.560 $\pm$ 0.027	3.10e4 $\pm$ 6.18e3	0.574 $\pm$ 0.030	6
	CL List	0.837	1.39e5	0.238	6
	CL Text	0.842	1.09e5	0.302	6
	CL LLM	0.701	1.85e4	0.413	3
	GT-base-2k List	0.745	1.56e4	0.320	4
	GT-base-2k Text	0.659	1.22e4	0.464	2
	GT-base-2k LLM	0.575	7.82e3	0.444	5
AHC	LF List	0.736	2.10e5	0.329	8
	LF Text	0.732	5.95e4	0.343	7
	LF LLM	0.553	9.83e3	0.407	5
	CL List	0.791	1.88e5	0.382	6
	CL Text	0.765	9.36e4	0.448	6
	CL LLM	0.647	3.49e4	0.521	6
	GT-base-2k List	0.725	6.91e4	0.546	6
	GT-base-2k Text	0.699	5.71e4	0.488	6
	GT-base-2k LLM	0.508	3.08e4	0.658	6
	LF List	<u>0.791</u>	<u>1.48e5</u>	0.373	6
	LF Text	0.748	7.28e4	0.444	6
	LF LLM	0.694	4.79e4	0.503	6

Best performance for each algorithm shown in bold; second-best underlined

Density-based clustering algorithms achieved the strongest performance

- HDBSCAN with the Clinical-Longformer List template embeddings was selected for downstream subgroup analysis and phenotyping (Fig. 2)

Manual templates outperformed decoder-generated summaries

- Decoder-LLM may omit or distort structured medication-use information

Medication-use phenotypes in postoperative delirium

Embedding-derived medication-use phenotypes correspond to distinct therapeutic strategies and recovery trajectories

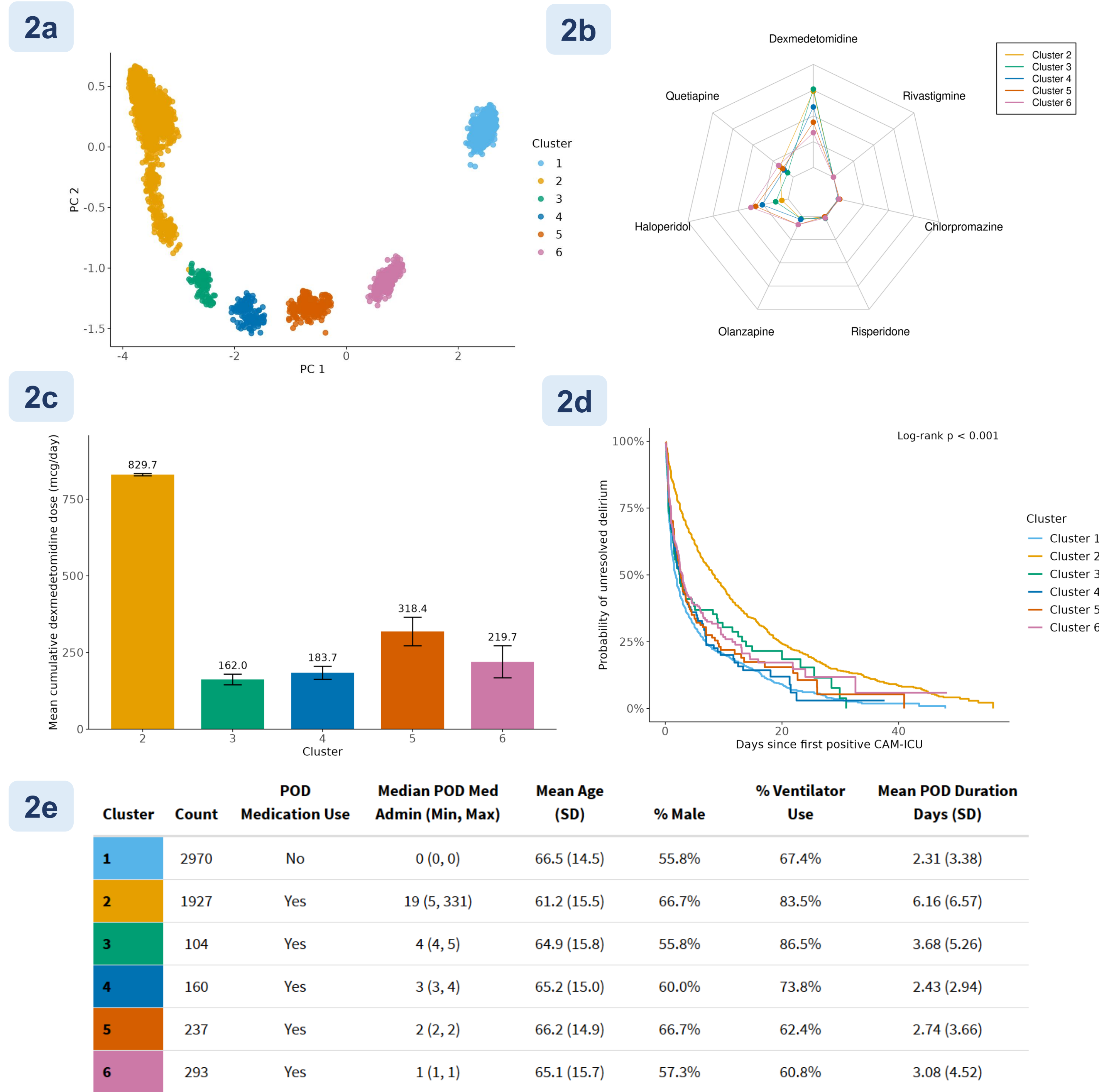


Figure 2. Clinical characteristics of the POD medication-use phenotypes identified by MEDLEY
2a) 2-dimensional PCA visualization of the clusters; 2b) Radar plot for POD medication administration proportions by cluster; 2c) Mean cumulative daily dexmedetomidine dose among patients with at least one dexmedetomidine administration; 2d) Kaplan-Meier curves showing the probability of unresolved delirium over time since the first positive delirium assessment; 2e) Cluster summary table

Conclusions

MEDLEY provides a reproducible representation learning framework for irregular longitudinal medication data

- Clinically pretrained LLMs capture complex medication-use trajectories via self-attention-based sequence modeling

Embedding-derived phenotypes were clinically coherent

- Subgroups with prolonged delirium exhibited greater relative sedative use, whereas shorter-duration subgroups showed higher haloperidol use

Interpretability of high-dimensional embeddings remains limited

- Future work will investigate attention-based visualization and feature attribution methods to quantify medication-level contributions

Key References

- [1] Li Y et al. *Clinical-Longformer and Clinical-BigBird: Transformers for long clinical sequences*. arXiv. 2022.
- [2] Yang X et al. *A large language model for electronic health records*. npj Digit Med. 2022.
- [3] Hegselmann S et al. *TabLLM: Few-shot classification of tabular data with large language models*. AISTATS. 2023.
- [4] Johnson AEW et al. *MIMIC-IV: A freely accessible electronic health record dataset*. Sci Data. 2023.